

**Procedure 16****Prepared by:**
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Clive Baek**Review Date:** March 2026**Original Date:** June 2018**Guidelines**

UBC Building Operations has developed a list of operational assessments that are to be carried out by Operating Engineers at the CCM twice in every 12 hours shift. The first set of rounds/checks are to be carried as soon as reasonably practical upon assuming of the shift, as per PEBPVRSR Section 11.

In all cases the rule of thumb to understand is that the **entire boiler room/mechanical room** is the “realm” of the Operating Engineer and any anomalies when performing your daily assessments must be reported within the operator’s log book entry.

- Assessment of the plant is required twice in a 12 hour shift. The first assessment must commence **within the first hour** of the shift.
- Perform a chemical treatment analysis and add chemicals to the boiler feed water accordingly during the first assessment.
- Assess all Technical Safety BC (TSBC) BPVR equipment in the facility during the second visit to the facility
- Enter all data in the log book as per check list provided and sign the same.
- Take a gas meter reading once per shift and record in log book
- **Ensure that the remote monitoring is active in the (Operations Centre) by taking the three steps as outlined below:**
 - **Contact the Operations Centre and ensure that the DDC graphics are updating and matching the CCM graphics**
 - **Trip the boiler (silence but don’t acknowledge alarm) and wait for confirmation from the Operations Centre that the strobe has been activated.**
 - **Have the Operations Centre confirm that the camera system is active and updating.**
- If any of the above remote monitoring checks fail, then the CCM Plant must be manned 24/7 with a minimum 4th Class Power Engineer until the remote monitoring issue is resolved. A call must be made to the Chief Engineer of the plant to arrange the appropriate coverage if needed as per call out procedure.
- DO NOT change any set point parameters without the prior approval of the People and Process Manager, Mechanical Operations.
- All abnormal conditions at the CCM Plant must be made safe, logged in the log book and an email communication sent to the Chief Engineer of the plant. This is for items such as items including steam leaks, feed water/condensate leaks, equipment vibration, etc.

Procedure

Check the following at least twice per shift:

- Review the general condition of the boiler room/mechanical room including any abnormalities, housekeeping, leaks, etc.
- (B1) Clayton Boiler and (B2) Cleaver Brooks boiler pressures (Normally 110-120 PSI)
- Steam header pressure (Normally 100-110 PSI)
- Stack temperature and record in log book
- Flame signals (normally 91 - 96%)
- Gas pressure (5 -7 PSI upstream of regulator, 1 PSI downstream)
- Boiler feed water pumps (check operation, noise, etc.)
- Water softeners integrator (counts down from 48,000 US gallons)
- Hot well temperature (normally 190 - 200°F and level)
- Log on to CCM Plant BMS workstation and assess all plant conditions
- Ensure hot water boilers are operating as intended and are running when CB boiler is in operation.
- Check steam vents to ensure that there isn't an abnormal amount of steam coming out

Check the following at least once per shift

- Blowdown hot well gauge glass
- Blowdown booster pump strainers
- Blowdown hot well float chamber
- Blowdown Mud Drum for Cleaver Brooks (Operating procedure # 8)
- Test manual LWCO devices (Operating procedure #15)
- Test boiler feed water, adjust rates, batch chemical and record results.
- Check oil level in Clayton feed water pump
- Record gas meter integrator (LSD in compound outside)
- Booster pump pressure (normally 10 – 20 PSI)
- HVAC systems, cooling tower, hot water systems, DHW heat exchangers, air compressors, steam traps

Note: Alarms require immediate attention. Header pressure decays rapidly and an outage of as little as 5 minutes could result in a loss of service for critical facilities operations. Response time is to be less than 30 minutes. Remote monitoring of the steam plant is from the Operations Centre.